TED (21) – 5091 **REVISION 2021**

FIFTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING AND TECHNOLOGY

TYRE TECHNOLOGY

MODEL QUESTION PAPER – SET-1

Time: 3 hours Maximum Marks: 75

PART A

I. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

		Module Outcome	Cognitive level
1	The stabilizer plies used in radial tyre is known as __________?	**M1.03**	U
2	Write down the expression for aspect ratio?	**M1.03**	U
3	Name the nylon cord materials used in tyre carcass?	**M1.04**	U
4	Name the tread pattern recommended for truck tyre back wheels?	**M2.01**	U
5	Flat spotting is happened due to the __________ property of thermoplastic tyre cords?	**M2.03**	U
6	In tubeless tyre inner liner is made up of __________ rubber??	**M2.04**	U
7	Name the process used to build cycle tyres from a spool of carcass strip?	**M3.01**	U
8	X-ray imaging is an example of ____________ test?	**M4.02**	U
9	Leak test of cycle tube is carried out at an inflation to get ______ % increase in tube's outer circumference?	**M4.04**	U

PART B

II. Answer any Eight questions from the following

(8 x 3= 24 Marks)

		Module Outcome	Cognitive level
1	Using a sketch mention the function of bead area components?	M 1.01	U
2	Write down the advantages and disadvantages of radial tyres?	**M1.02**	**U**
3	What is tyre designation? Give examples of bias and radial tyre size indication?	**M1.03**	**U**

4	Write down a typical RFL formulation used for Nylon dipping?	**M1.03**	**U**
5	Describe flat spotting and aqua planing?	**M2.02**	**U**
6	List the requirements of curing bladder?	**M2.03**	**U**
7	Using a neat sketch label the parts of a cycle tyre?	**M2.03**	**U**
8	Give a suitable recipe for cycle tyre inner tube based on NR?	**M3.01**	**U**
9	Write notes on infrared and ultrasonic tests on tyres?	**M3.02**	**U**
10	Outline the data collected through cut tyre analysis?	**M4.02**	**U**

PART C

III. Answer all questions from the following (6x 7 = 42 Marks)

		Module Outcome	Cognitive level
1	Describe different tyre constructions using sketches?	M1.03	**U**
	OR		
2	Describe the auxiliary materials used in tyre manufacture?	M1.04	**U**
3	Suggest a formulation with justification for passenger car tyre tread?	M2.01	**A**
	OR		
4	Narrate the adhesive treatment of polyester tyre cord using a sketch?	M2.02	U
5	Illustrate the production of carcass plies using a flow chart?	M2.01	U
	OR		
6	Discuss different types of curing presses used in tyre manufacture?	M2.03	U
7	Describe different methods used for cycle tube production?	M3.02	U
	OR		
8	Outline the methods used for the manufacture of industrial solid tyres?	M3.03	U
9	Account the stages in conventional re-treading of tyres?	M3.04	U
	OR		
10	Compare the hot and cold re-treading processes?	M3.04	U

11	Discuss the tyre non-uniformity situations in pneumatic tyres?	M4.02	U
OR			
12	Outline the tension set tests carried out on cycle tyres and cycle tubes?	M4.04	U

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FIFTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING AND TECHNOLOGY

TYRE TECHNOLOGY

MODEL QUESTION PAPER – SET-2

Time: 3 hours Maximum Marks: 75

PART A

IV. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

		Module Outcome	Cognitive level
1	Who is the inventor of successful pneumatic tyre ?	M1.01	R
2	The fabric part of a pneumatic tyre is known as _________?	M1.02	U
3	Low aspect ratio tyres are _______ in tyre width than tyre height?	M1.03	U
4	A __________dipping process is recommended for polyester cords?	M2.02	U
5	The cooling of tyre in inflated condition after removal from mould is known as _________?	M2.03	U
6	The process of joining two ends of inner tube is termed as _______?	M2.04	U
7	The projected side wall portion of cycle tyres are called _________?	M3.01	U
8	______ solution dipping is widely adapted in tyre industries for the identification of specific gravity of tyre compounds?	M4.01	U
9	_______ test is used for the bead wire strength checking of cycle tyres?	M4.04	U

PART B

V. Answer any Eight questions from the following

(8 x 3= 24 Marks)

		Module Outcome	Cognitive level
1	Point out the functions of a pneumatic tyre?	M1.02	U
2	Distinguish different tyre constructions depending on carcass and belt angles?	M1.03	U

3	Define aspect ratio and sketch a 60 aspect ratio tyre?	M1.03	U
4	What is meant by ply rating?	M2.02	U
5	Show the stages in the production of industrial solid tyres using a flow chart?	M3.03	U
6	What are the advantages and disadvantages of tubeless tyres?	M2.04	U
7	Describe different production methods used for cycle tubes?	M3.02	U
8	Give a suitable recipe for cushion gum compound used in re-treading?	M3.04	U
9	List different courses adapted in Proving Ground Test?	M4.03	U
10	Outline the Denison bend test?	M4.04	U

PART C

VI. Answer all questions from the following **(6x 7 = 42 Marks)**

		Module Outcome	Cognitive level
1	Distinguish radial and bias tyres?	M1.03	U
OR			
2	Write a note on tyre nomenclature/size designation?	M1.03	U
3	Narrate the adhesive treatment of Nylon tyre cord using a sketch?	M2.02	U
OR			
4	Suggest a formulation with justification for passenger car tyre tread?	M2.01	A
5	Describe different methods used for cycle tyre building?	M3.01	U
OR			
6	Compare solid tyres with pneumatic tyres?	M3.03	U
7	Describe the endurance test on pneumatic tyres?	M4.02	U
OR			
8	Outline the stages in highway test of field evaluation on pneumatic tyres?	M4.02	U
9	Narrate the stages in the building of a bias tyre?	M2.03	U
OR			

10	Illustrate the production of carcass plies using a flow chart?	M2.01	U
11	Account the stages in conventional re-treading of tyres?	M3.04	U
OR			
12	Compare the hot and cold re-treading processes?	M3.04	U

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Mark Distribution

Module	Hour/Module	Marks/Module (hi /ΣHi) * 123 (±5%)	Type of Questions							
			Part A		Part B		Part C		Total	
			No. of Questions	Marks	No. of Questions	Marks	No. of Questions	Marks	No. of Questions	Marks
I	15	33	3	3	3	9	2	14	8	26
II	16	33	3	3	3	9	4	28	10	40
III	14	28	1	1	2	6	4	28	7	35
IV	15	29	2	2	2	6	2	14	6	22
Total	60	123	9	9	10	30	12	84	31	123

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Cognitive Level Mark Distribution

Cognitive Level	Marks	% of Marks
Remembering	- -	- -
Understanding	113	91.87
Applying	10	8.13
Total	123	100

Question Wise Analysis

Qn. No.	Course Outcome	Cognitive Level	Marks	Time
I.1	M1.03	Understanding	1	2
I.2	M1.03	Understanding	1	2
I.3	M1.04	Understanding	1	2
I.4	M2.01	Understanding	1	2
I.5	M2.03	Understanding	1	2
I.6	M2.04	Understanding	1	2
I.7	M3.01	Understanding	1	2
I.8	M4.02	Understanding	1	2
I.9	M4.04	Understanding	1	2
II.1	M1.02	Understanding	3	7
II.2	M1.03	Understanding	3	7
II.3	M1.03	Understanding	3	7
II.4	M2.02	Understanding	3	7
II.5	M2.03	Understanding	3	7
II.6	M2.03	Understanding	3	7
II.7	M3.01	Understanding	3	7
II.8	M3.02	Applying	3	7
II.9	M4.02	Understanding	3	7
II.10	M4.02	Understanding	3	7
III.1	M1.03	Understanding	7	17
2	M1.04	Understanding	7	17
3	M2.02	Applying	7	17

4	**M2.02**	**Understanding**	**7**	**17**
5	**M2.01**	**Understanding**	**7**	**17**
6	**M2.03**	**Understanding**	**7**	**17**
7	**M3.02**	**Understanding**	**7**	**17**
8	**M3.03**	**Understanding**	**7**	**17**
9	**M3.04**	**Understanding**	**7**	**17**
10	**M3.04**	**Understanding**	**7**	**17**
11	**M4.02**	**Understanding**	**7**	**17**
12	**M4.04**	**Understanding**	**7**	**17**